

FAKE SOCIAL MEDIA ACCOUNT DETECTION

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Abstract:

The rapid expansion of social media platforms has resulted in increased misuse through the creation of fake or bot accounts. These accounts often spread misinformation, perform fraudulent activities, or manipulate public opinion. This research aims to build a machine-learning-based model capable of detecting fake social media accounts using profile-level and activity-based features. A dataset consisting of labelled *real* and *fake* accounts was used to train multiple classification algorithms. The best-performing model demonstrated strong predictive capability with high accuracy and robust generalization. The study highlights the essential features influencing detection and provides a scalable approach for automated moderation on online platforms.

1. Introduction

Social media has become a primary mode of communication, commerce, and information exchange. However, the rise of fake accounts both automated (bots) and human-operated poses significant threats. These accounts are responsible for misinformation campaigns, fake engagement, phishing attempts, and scams. As the scale of social media grows, manual detection becomes impractical, creating demand for robust automated systems.

The objective of this research is to design and evaluate a machine learning model that reliably distinguishes fake accounts from genuine users by analyzing behavioral and profile-related features. Using a labelled dataset, we explore supervised learning techniques and evaluate their effectiveness.

2. Literature Review

Rule-based and heuristic systems that employed basic behavioural indicators like odd follower-following ratios, excessive posting activity, default profile images, and randomly generated

usernames were the mainstay of early attempts to detect fake accounts on social media. While these manual and heuristic approaches were helpful in identifying low-quality spam accounts, some studies showed that they lacked adaptability as fake accounts evolved to more closely resemble human behaviour. Researchers started using supervised machine learning techniques as labelled datasets grew. Based on metadata and behavioural characteristics, Logistic Regression, XGBoost, and Decision Trees have become popular tools for differentiating human accounts from bots.

3. Dataset Description

The dataset used in this study consists of labelled data representing attributes of social media accounts, including:

- **Profile features:** username length, bio length, verification status
- **Activity features:** number of posts, followers, following
- **Engagement:** like/follower ratio, comment/post ratio
- **Account behaviour flags:** presence of profile picture, sudden spikes in activity, etc.

The dataset contains X samples with Y real and Z fake accounts. Basic preprocessing was applied, including handling missing values, normalization, and label encoding.

4. Methodology

The methodology of this study consists of data preprocessing, feature selection, training of three supervised learning models **Decision Tree**, **Random Forest**, **Logistic Regression** and XGBoost followed by performance evaluation

4.1 Data Preprocessing

Before training, the dataset underwent several preparation steps to ensure consistency and optimal performance:

4.1.1 Handling Missing Values

Missing values were either replaced with median values or converted into binary indicators (e.g., “bio missing = 1”).

4.1.2 Encoding Categorical Features

- Binary values like `profile_picture_present` and `is_verified` were label-encoded (0 or 1).
- No one-hot encoding was needed as dataset contained mostly numerical attributes.

4.1.3 Feature Scaling

- Logistic Regression requires feature scaling, so numerical features were normalized using `StandardScaler`.
- Decision Tree and Random Forest do not require scaling, but scaled data does not affect performance negatively, so the same normalized dataset was used for all models for uniformity.

4.1.4 Train-Test Split

- The dataset was split into:
 - **80% training data**
 - **20% testing data**
- Stratified splitting ensured equal distribution of fake and real accounts in both sets.

4.2 Feature Selection

To determine the most influential predictors the following methods are used:

- Methods Used
 - **Correlation Matrix:** Identified strong relationships between features and the target label.
 - **Feature Importance from Random Forest:** Used as a reliable indicator of non-linear feature impact.
 - **Domain Knowledge:** Social media bot studies highlight follower patterns, activity frequency, and profile completeness as significant factors.
- High-Impact Features Identified
 - Follower-following ratio
 - Number of posts
 - Profile picture presence
 - Account age
 - Average engagement rate

These features served as strong semantic indicators of fake behaviour.

4.3 Models Used

This study compares **three widely used supervised classification algorithms** to determine the most effective for detecting fake social media accounts.

4.3.1 Logistic Regression

Logistic Regression is a linear classification model that predicts the probability of an account being fake.

- **Advantages of Logistic Regression**
 - Fast to train
 - Easily interpretable
 - Performs well with linearly separable data
 - Provides insight into positive/negative influence of features
- **Model Training Parameters**
 - Test_size = 0.2
 - Random_state = 42

- `Max_iter = 80000`
- `Stratify = y`
- **Strengths**
 - Simple, fast, and easy to implement
 - Highly interpretable — feature impact is easy to understand
 - Performs well on linearly separable data
 - Less prone to overfitting when regularization is applied
- **Limitations**
 - Struggles with non-linear relationships in data
 - Performance decreases on complex or high-variance datasets
 - Assumes independence between features, which may not hold
 - Requires proper feature scaling and preprocessing to perform well

4.3.2 Decision Tree Classifier

A decision tree is a hierarchical structure that splits data based on feature thresholds.

- **Advantages of Decision Tree**
 - Handles non-linear patterns
 - Easy to visualize and interpret
 - Works without feature scaling
 - Captures interactions between features
- **Training Parameters**
 - Criterion: Gini impurity (default)
 - Max depth: 12
 - Min samples split: Adjusted to control tree complexity
- **Strengths**
 - Handles non-linear and complex relationships effectively
 - Easy to visualize and interpret
 - Requires no feature scaling or normalization
 - Captures interactions between features naturally
- **Limitations**
 - Highly prone to overfitting on training data
 - Sensitive to small changes in data (unstable)
 - Can create overly complex trees without pruning
 - Lower generalization compared to ensemble methods

4.3.3 Random Forest Classifier

Random Forest combines multiple decision trees using bootstrap aggregation.

- **Why Random Forest?**

- Reduces overfitting
- Handles high-dimensional data well
- Provides strong generalization
- Offers reliable feature importance scoring
- Captures complex non-linear interactions
- **Training Parameters**
 - `N_estimators = 350`
 - `max_depth = 20,`
 - `random_state = 42,`
 - `n_jobs = -1`
- **Strengths**
 - Reduces overfitting by averaging multiple decision trees
 - Handles high-dimensional and complex data effectively
 - Provides strong generalization and high accuracy
 - Offers reliable feature importance insights
- **Limitations**
 - More computationally expensive than a single decision tree
 - Less interpretable due to multiple trees in the ensemble
 - Can become slow with a very large number of trees
 - May still overfit if trees are very deep or poorly tuned

4.3.4 XGBoost

XGBoost (Extreme Gradient Boosting) is an optimized boosting algorithm that builds an ensemble of decision trees sequentially, each correcting the errors of the previous ones.

- **Why XGBoost?**
 - Handles complex non-linear patterns
 - Highly accurate due to boosting
 - Built-in regularization reduces overfitting
 - Efficient and fast even on large datasets
 - Manages missing values automatically
 - Performs well with imbalanced data
- **Training Parameters**
 - `n_estimators: 300`
 - `max_depth: 8`
 - `learning_rate : 0.08`
 - `subsample: 0.9`
 - `colsample_bytree: 0.9`
 - `n_jobs = -1`
 - `verbosity = 0`
- **Strengths**
 - Often achieves higher accuracy than single trees

- Excellent generalization with regularization
- Robust to noise and feature interactions
- Tunable for class imbalance scenarios
- **Limitations**
 - More complex and less interpretable than a single tree
 - Requires careful hyperparameter tuning
 - Computationally heavier than Logistic Regression
 - Can overfit if trees are too deep or learning rate is too high

5. Experiments and Results

All three models were trained on the same processed dataset and evaluated using standard metrics.

5.1 Evaluation Metrics

To fully understand model performance, the following metrics were calculated:

- **Accuracy:** Overall correctness
- **Precision:** Reliability of detecting fake accounts
- **Recall:** Ability to identify all fake accounts
- **F1-Score:** Harmonic balance between precision and recall
- **Confusion Matrix:** Breakdown of correct vs. incorrect predictions

Since fake accounts are harmful, recall is particularly important missing a fake account (false negative) is costlier than misclassifying a real one.

5.2 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	94.89 %	0.95	0.94	0.95
Decision Tree	96.51%	0.96	0.97	0.97
Random Forest	96.68%	0.96	0.97	0.97
XGBoost				

5.3 Interpretation of Results

- Logistic Regression:
 - Performed decently but underfitted the complex patterns of fake accounts.
 - Linear decision boundary was insufficient for high-variance dataset.

- Decision Tree:
 - Captured non-linear behavior effectively.
 - Slight overfitting was observed, but performance remained strong.
- Random Forest:
 - Best overall performance due to ensemble learning.
 - Balanced precision and recall makes it ideal for real deployments.
 - Reduced overfitting compared to decision tree.

5.4 Feature Importance Analysis (Random Forest Based)

The top predictive features includes:

- Follower/following ratio
- Number of posts
- Account age
- Profile image presence
- Engagement rate

These align with established indicators of bot-like behaviour found in prior research.

6. Conclusion

The findings of this research offer proof that machine learning algorithms can successfully differentiate genuine social media accounts from fraudulent ones by analysing behavioural and profile-related characteristics. Of the three algorithms evaluated Logistic Regression, Decision Tree and Random Forest distinct variations, in performance were observed. This underscores the significance of model complexity. How feature interactions impact classification effectiveness.

Logistic Regression served as an transparent baseline approach. Its linear decision boundary enabled it to identify trends within the data. Nevertheless it found it difficult to handle the non-linear patterns frequently associated with fraudulent account activities. As a result its accuracy and F1-score were comparatively reduced. Although it proves valuable for interpreting the influence and effect of features Logistic Regression falls short when applied to highly diverse datasets such, as social media account metadata.

The Decision Tree classifier demonstrated performance by capturing non-linear relationships and interactions, between features. Its tree-like structure allowed it to pinpoint behavioral characteristics, including follower-following ratio, account age and posting frequency. Nonetheless Decision Trees tend to overfit particularly when dealing with imbalanced datasets. While the model yielded outcomes its generalization capability was restricted compared to ensemble approaches.

The Random Forest model achieved the overall results across all assessment criteria. By aggregating decision trees through bagging it minimized overfitting, enhanced robustness and identified more intricate feature relationships. The model demonstrated precision and recall

making it highly effective for practical fake account identification, where failing to detect fake accounts poses a greater risk, than incorrectly flagging legitimate accounts. The evaluation of feature importance, by Random Forest likewise provided understanding revealing that behavioral trends, account completeness and engagement measures serve as the primary predictors of account legitimacy.

In summary, this research successfully established a machine learning framework that can accurately identify fake accounts. The methodology is scalable, interpretable, and can be applied to various platforms. This work enhances the safety and integrity of online communities bproviding an automated solution to help social media platforms in fraud detection and moderation efforts.

7. References

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